

Multigrid Finite State Projection for the Reaction-Diffusion Master Equation

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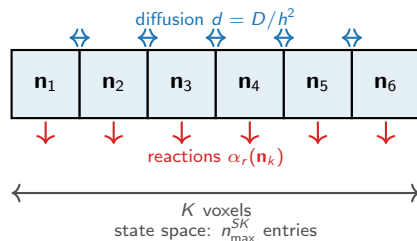
The RDME: reactions and diffusion across a spatial grid

Model. Divide a 1-D domain into K voxels. Each voxel k holds molecule counts $\mathbf{n}_k \in \mathbb{Z}_{\geq 0}^S$.

Two types of events:

- **Reactions** inside voxel k : rate $\alpha_r(\mathbf{n}_k)$, change $\mathbf{n}_k$.
- **Diffusion** between neighbors: rate $d = D/h^2$, moves one molecule $k \leftrightarrow k+1$.

The joint probability $p(\mathbf{n}_1, \dots, \mathbf{n}_K, t)$ satisfies $\dot{p} = Ap$ (**RDME**).



The computational bottleneck

The state space has $\sim n_{\max}^{SK}$ reachable states — **exponential in K** . Even $K=8, S=1$ is intractable with a naive FSP.

$\dot{p} = Ap$ has **block-tridiagonal structure** (schematic, $K=4$, $S=1$)

each block: $n_{\max}^S \times n_{\max}^S$ total: $n_{\max}^{SK} \times n_{\max}^{SK}$

A_k : reactions inside voxel k
voxel k

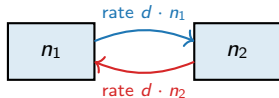
$D_{k,k+1}$: diffusion between neighboring voxels

p_k : probability sub-vector for

The Binomial is the exact diffusion equilibrium

Fix total molecules $\bar{n} = n_1 + n_2$ across two adjacent voxels.

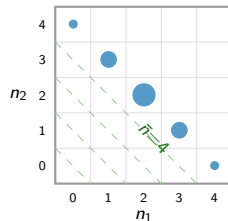
Diffusion moves one molecule at a time, at rate d per molecule:



Each molecule hops independently $\Rightarrow$ at equilibrium, equally likely to be in either voxel. For $\bar{n}$ total:

$$P(n_1 \mid \bar{n}) = \text{Bin}(\bar{n}, \frac{1}{2}) \quad (\text{exact, any } d > 0)$$

When the distribution is near this equilibrium, $\bar{n}$ **alone describes the two-voxel state.**



Probability spreads along constant-total diagonals as $\text{Bin}(\bar{n}, \frac{1}{2})$ (dot size $\propto$ mass).

The two-voxel state is captured by $\bar{n}$ alone.

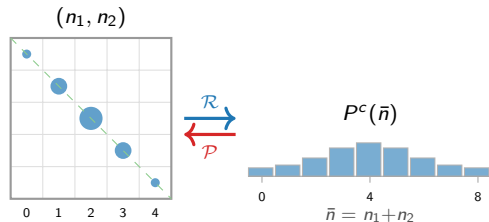
Merging two adjacent voxels: restriction and prolongation

Compress ($\mathcal{R}$): exact:

$$p^c(\bar{n}) = \sum_{n_1+n_2=\bar{n}} p(n_1, n_2).$$

Reconstruct ($\mathcal{P}$): approximate:

$$p(n_1, n_2) \approx p^c(\bar{n}) \cdot \text{Bin}(\bar{n}, \frac{1}{2}).$$



State-space reduction

Per pair: $n_{\max}^{2S} \rightarrow n_{\max}^S$.

All K voxels: $n_{\max}^{SK} \rightarrow n_{\max}^{SK/2}$.

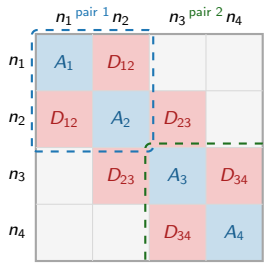
$\mathcal{P}$ is accurate when the distribution is near-Binomial

$\mathcal{R}$: sum along each diagonal $\rightarrow$ scalar $\bar{n} = n_1 + n_2$.

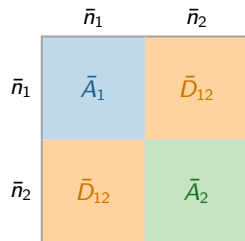
$\mathcal{P}$: spread back via $\text{Bin}(\bar{n}, \frac{1}{2})$.

Coarsening the RDME Matrix

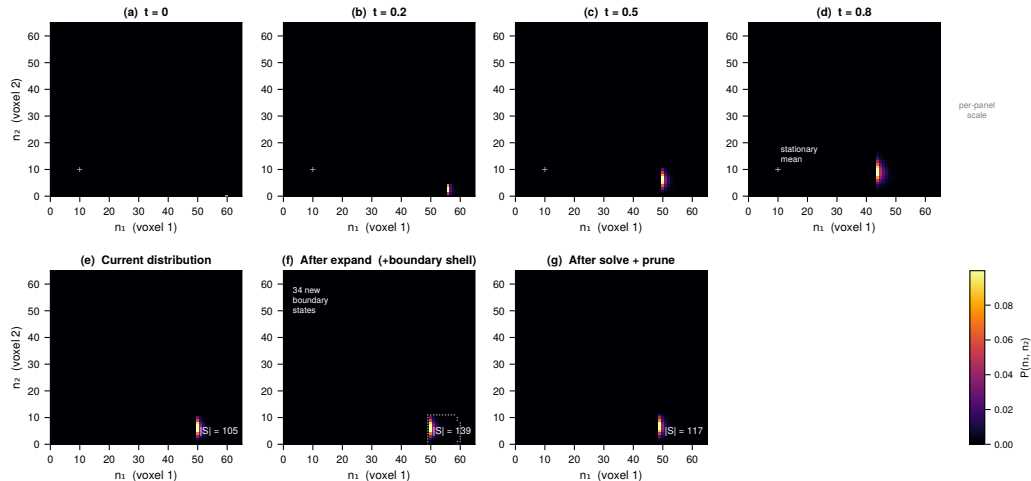
Full matrix A : 4 voxels:



Merged matrix $\bar{A}$: 2 pairs:

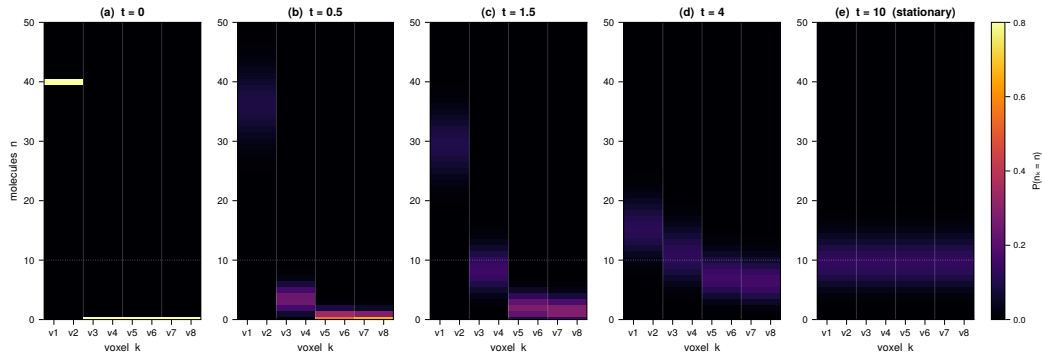


The adaptive FSP algorithm: expand, solve, prune



Top row: joint distribution $P(n_1, n_2)$ of a 2-voxel birth-death system: point source at $(n_{src}, 0)$ traveling to the stationary point (white cross).

Result: K=8 RDME wavefront via coarsened simulation



K=8 wavefront solved via K=4 coarse FSP. Max coarse state space: 106585 states. K=8 direct FSP would be intractable (grows to 90k+ states by $t=0.8$). Fine marginals recovered analytically via $\text{Binom}(n_k, \frac{1}{2})$ prolongation — exact for balanced pairs.

Birth-death RDME, $K=8$, $D=0.1$. IC: $v_1=v_2=40$ (delta), $v_3 \dots v_8=0$. Each panel shows marginals $P(n_k=n)$ for all 8 voxels.

(a)–(d): Wavefront sweeps left→right, distributions broaden. **(e)**: Stationary: all voxels reach Poisson(10) (dashed line).

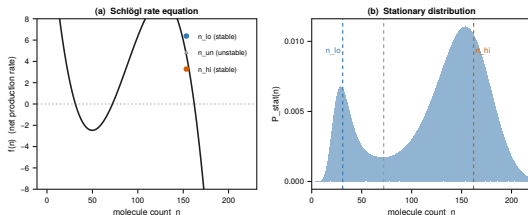
Solved via **K=4 coarse FSP** (30k states). Direct K=8 FSP: intractable. Fine marginals recovered via $\text{Bin}(\tilde{n}, \frac{1}{2})$:

Bistable reactions

For birth-death, each voxel has one stable state.
Diffusion wins: the pair is always well-mixed.
Binomial coarsening is *essentially exact*.

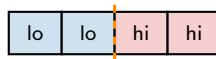
But what if reactions are bistable?

The **Schlögl model** has two stable states per voxel:
 $n_{lo} \approx 31$ and $n_{hi} \approx 162$.



Spatial metastability

Neighbors can occupy *different* states. A **spatial front** forms.



Reactions “pull” voxels apart; diffusion “pulls” them together. When reactions are fast, the front **persists for very long times**.

Why Binomial prolongation fails at bistable fronts

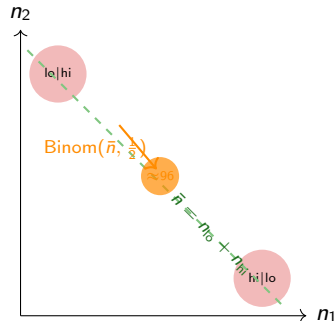
At a spatial interface, the joint distribution $P(n_1, n_2)$ is **bimodal**: mass concentrated at (n_{lo}, n_{hi}) and (n_{hi}, n_{lo}) .

Restriction loses orientation

$$\mathcal{R}(n_{lo}, n_{hi}) = n_{lo} + n_{hi} = \mathcal{R}(n_{hi}, n_{lo}).$$

The total $\bar{n} = 193$ is the same whether the low-state voxel is on the left or the right. **The front direction is erased.**

Binomial prolongation then places the reconstructed mass at $(n_{hi}/2 + n_{lo}/2, \dots) \approx (96, 97)$ — squarely in the **unstable** region. The metastable front collapses to an unphysical average.



$\mathcal{P}$ spreads mass along the diagonal via $\text{Binom}(\bar{n}, \frac{1}{2})$, placing it at the **unstable fixed point** instead of preserving the two bistable modes.

The n -level recursive V-cycle

One V-cycle step at level K :

- 1 **Pre-smooth**: evolve intra-pair diffusion for $\tau_{\text{pre}} \rightarrow$ drives within-pair distribution toward $\text{Bin}(\bar{n}_j, \frac{1}{2})$
- 2 **Restrict**: $\bar{n}_j = n_{2j-1} + n_{2j}$; save $\bar{p}^{\text{pre}} = \mathcal{R} \tilde{p}$ (pre-solve coarse distribution)
- 3 **Recurse**: V-cycle on $K/2$ -voxel system (direct FSP at the coarsest level $K/2^L$)
- 4 **Prolong**: conditional reconstruction from coarse update $\bar{p}^{\text{post}}$
- 5 **Prune**: remove states with $p(x) < \tau$

Hierarchy

$$K \rightarrow K/2 \rightarrow K/4 \rightarrow \dots \rightarrow K/2^L$$

Coarsest level solved exactly with the FSP. Each level adds one restriction + prolongation step.

State space

Grows *only* through prolongation — no explicit enumeration of the full n_{max}^K space. Support of p tracked directly; controlled by the prune threshold τ .

Conditional prolongation: reconstructing the fine distribution

After the coarse solve, $\bar{p}^{\text{post}}$ differs from $\bar{p}^{\text{pre}}$. Three types of coarse states arise:

1. Covered states ($\bar{x} \in \mathcal{S}^{\text{pre}} = \mathcal{R} \tilde{p}$): multiplicative update

$$p^{\text{new}}(x) = \tilde{p}(x) \cdot \frac{\bar{p}^{\text{post}}(\bar{x})}{\bar{p}^{\text{pre}}(\bar{x})}.$$

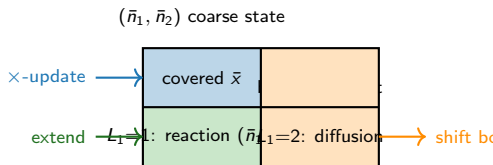
Exact and asymmetry-preserving: mass at $(n_{\text{lo}}, n_{\text{hi}})$ stays there rather than being forced symmetric.

2. Reaction boundary ($L_1 = 1$: one pair count changes by ± 1):

Extend the nearest covered conditional — keep n_{2j-1} , set $n_{2j} = \bar{n}_j^{\text{new}} - n_{2j-1}$.

3. Diffusion boundary ($L_1 = 2$: two pair counts each change by ± 1):

Coarse diffusion moves a molecule between pairs simultaneously. Shift second voxels of *both* changed pairs.

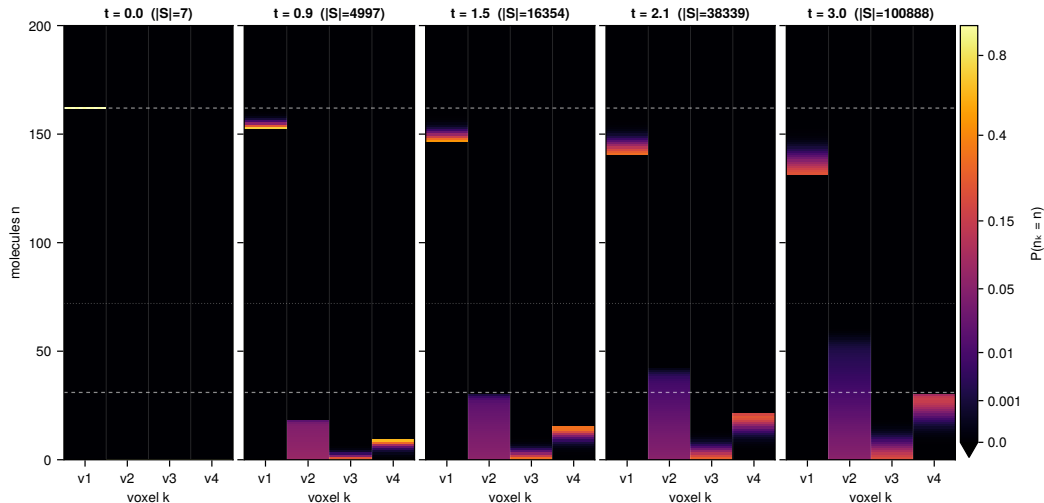


Why $L_1=2$ matters

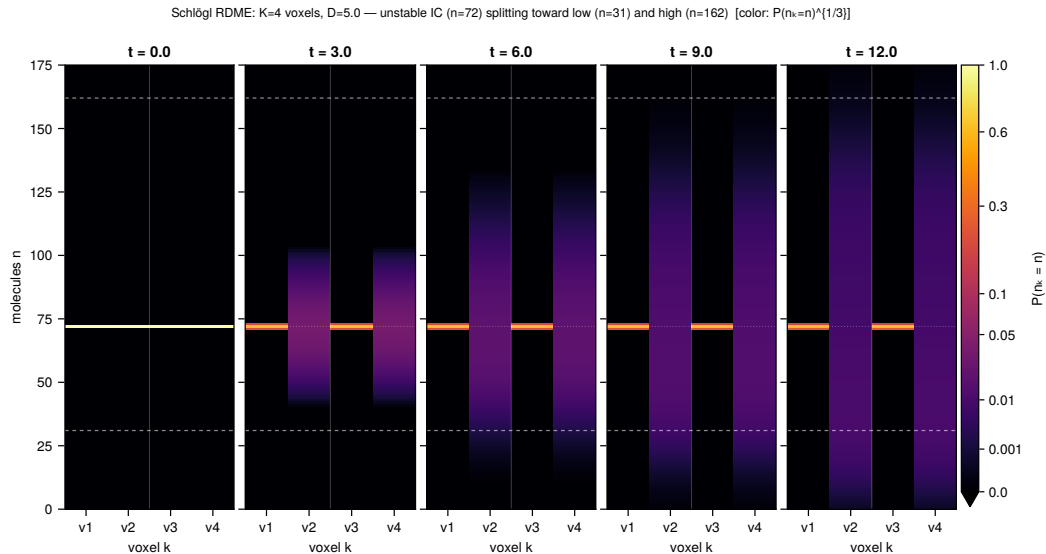
Coarse diffusion $(\bar{n}_1, \bar{n}_2) \rightarrow (\bar{n}_1-1, \bar{n}_2+1)$ is always $L_1=2$. Without this search, **inter-pair mass transport is invisible** to the V-cycle and molecules never leave pair 1.

Result: Schlögl wavefront spreading via the V-cycle

Schlögl RDME: $K=4$, $D=1.0$ — IC: all 162 molecules in v_1 , spreading via diffusion [color: $P^{1/3}$]



Result: splitting from the unstable Schlögl fixed point



Summary

1. Fast diffusion enables state-space compression via restriction.

When the within-pair distribution is near-Binomial, marginalising pairs reduces the state space from $n_{\max}^{2S}$ to $n_{\max}^S$ per pair. $K=8$ birth-death wavefront solved with 30k states (direct: intractable).

2. Bistable systems require conditional — not Binomial — prolongation.

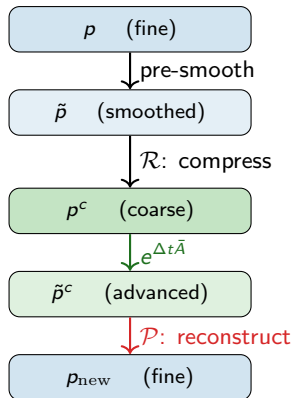
At spatial fronts, the within-pair distribution is bimodal. Conditional prolongation preserves asymmetry: multiplicative update for covered states; $L_1 \leq 2$ neighbor search handles both reaction ($L_1=1$) and diffusion ($L_1=2$) boundary states.

3. The n -level V-cycle propagates correct spatial dynamics.

Wavefront spreading (from concentrated IC) and bistable splitting (from unstable IC) are both captured. State-space growth is controlled by pruning: no explosion.

Open questions: error bounds on conditional prolongation; dynamic- π for improved within-pair accuracy; multi-species Schlögl systems.

Appendix: why not reconstruct at every step?

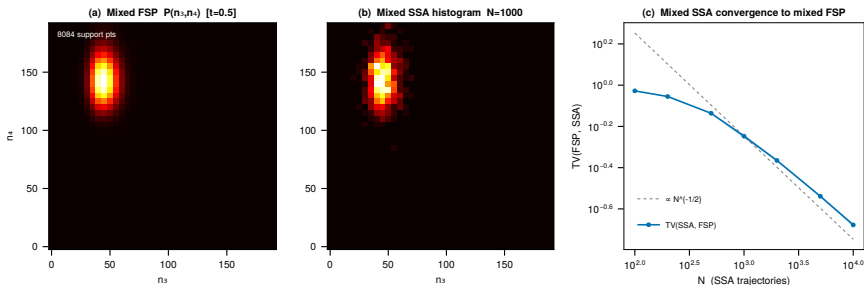


Reconstruction is wrong at the front

Step 4 forces a Binomial split on *every* pair, including the front where the true distribution is bimodal.

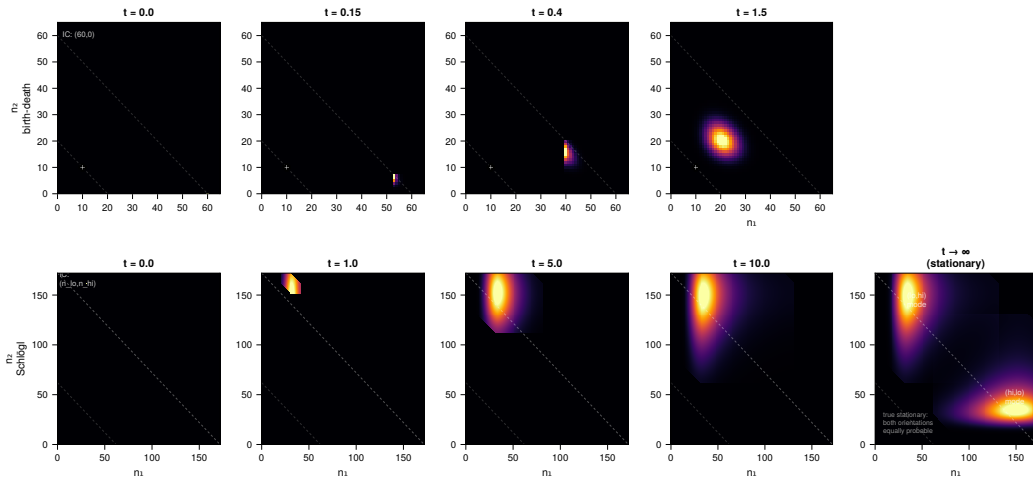
Approach: evolve p^M directly: no reconstruction step. Coarse pairs carry the Binomial assumption implicitly in $\bar{A}_j$; the fine window tracks the true within-pair distribution exactly.

Appendix: mixed-resolution generator is self-consistent



ODE solve on the mixed state $(\bar{n}_1, n_3, n_4)$ vs. Monte Carlo on the same reduced generator. Means agree to $<1\%$; TV $\sim N^{-1/2}$ (sampling noise only).

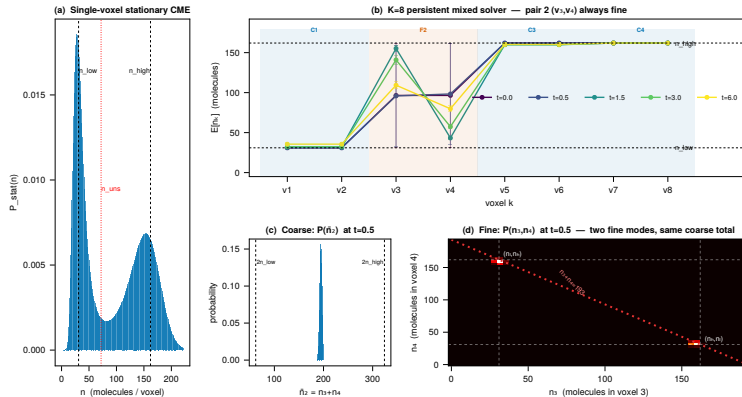
Appendix: probability dynamics (birth-death vs. Schlögl)



Top: birth-death point source traveling to Poisson(10) stationary. Binomial coarsening exact.

Bottom: Schlögl front IC persisting metastably; true stationary (rightmost) shows both orientations equally.

Appendix: K=8 persistent front (detailed)



Appendix: K=12 persistent front

